

Problems (Advanced Control)

[MPC]

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1. We take another look at Example 6.1 from lectures.

Suppose that $x(k) \in \mathbb{R}^2$ and $u(k) \in \mathbb{R}$ satisfy

$$x(k+1) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} x(k) + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u(k)$$

and that MPC is applied with the state and control penalties

$$Q = \begin{pmatrix} 1 & 2 \\ 2 & 6 \end{pmatrix}, \quad R = \frac{1}{2}.$$

Now take the horizon length as $H = 2$.

- (i) Show that the MPC cost function

$$V_2(x(k)) = \sum_{i=0}^2 \|\hat{x}(k+i|k)\|_Q^2 + \sum_{i=0}^1 \|u(k+i|k)\|_R^2$$

can be written as

$$V_2(x(k)) = \frac{15}{2}u_1^2 + 4u_1u_2 + \frac{3}{2}u_2^2 + 4u_1x_1 + 6x_1^2 + \|x(k)\|_Q^2$$

where $u_1 = u(k|k)$ and $u_2 = u(k+1|k)$ and x_1 is the first element of $x(k)$.

- (ii) Show that

$$\frac{\partial V_2}{\partial u_1} = 15u_1 + 4u_2 + 4x_1, \quad \frac{\partial V_2}{\partial u_2} = 4u_1 + 3u_2.$$

- (iii) Solve the simultaneous equations

$$\frac{\partial V_2}{\partial u_1} = 0, \quad \frac{\partial V_2}{\partial u_2} = 0$$

and show that the optimal control inputs satisfy

$$u^*(k|k) = -\frac{12}{29}x_1, \quad u^*(k+1|k) = -\frac{4}{3}u^*(k|k).$$

Infer that the MPC feedback law $\kappa : \{\text{states}\} \mapsto \{\text{inputs}\}$ is given by

$$\kappa(x) = K_2x = \begin{pmatrix} -\frac{12}{29} & 0 \end{pmatrix} x.$$

- (iv) Show that the closed-loop MPC dynamics $x(k+1) = (A + BK_2)x$ are stable.

$$(i) \quad \|x(0)\|_a^2 = [x_1 \ x_2] \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = [x_1 + 2x_2 \quad 2x_1 + 6x_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + 4x_1x_2 + 6x_2^2$$

$$\|u(0)\|_a^2 = u_1 \cdot \frac{1}{2} u_1 = \frac{1}{2} u_1^2$$

$$x(1) = \begin{bmatrix} 0 \\ x_1 \end{bmatrix} + \begin{bmatrix} u_1 \\ 0 \end{bmatrix} = \begin{bmatrix} u_1 \\ x_1 \end{bmatrix}$$

$$\therefore \|x(1)\|_a^2 = [u_1 \ x_1] \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} u_1 \\ x_1 \end{bmatrix} = [u_1 + 2x_1 \quad 2u_1 + 6x_1] \begin{bmatrix} u_1 \\ x_1 \end{bmatrix} = u_1^2 + 4u_1x_1 + 6x_1^2$$

$$\|u(1)\|_a^2 = \frac{1}{2} u_1^2$$

$$x(2) = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ u_1 \end{bmatrix} + \begin{bmatrix} u_2 \\ 0 \end{bmatrix} = \begin{bmatrix} u_2 \\ u_1 \end{bmatrix}$$

$$\therefore \|x(2)\|_a^2 = [u_2 \ u_1] \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} u_2 \\ u_1 \end{bmatrix} = u_2^2 + 4u_2u_1 + 6u_1^2$$

$$\therefore V_2(x(k)) = \frac{15}{2} u_1^2 + 4u_2u_1 + \frac{3}{2} u_2^2 + 4u_1x_1 + 6x_1^2 + \|x(k)\|_a^2$$

$$\text{where } \|x(k)\|_a^2 = x_1^2 + 4x_1x_2 + 6x_2^2$$

$$(ii) \quad \frac{\partial V_2}{\partial u_1} = 15u_1 + 4u_2 + 4x_1$$

$$\frac{\partial V_2}{\partial u_2} = 4u_1 + 3u_2$$

$$(iii) \quad \text{for } 15u_1^* + 4u_2^* + 4x_1 = 4u_1^* + 3u_2^* = 0 \quad \text{at } u = u^*$$

$$u_1^* = -\frac{3}{4} u_2^* \quad \therefore -\frac{29}{4} u_2^* + 4x_1 = 0$$

$$\therefore u_2^* = -\frac{16}{29} x_1 = u^*(k+1|k) \quad \therefore u_1^* = -\frac{12}{29} x_1 = u^*(k|k)$$

$$(iv) \quad A + BK_2 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} -\frac{12}{29} & 0 \end{bmatrix} = \begin{bmatrix} -\frac{12}{29} & 0 \\ 1 & 0 \end{bmatrix}$$

$$\therefore \det \{ (A + BK_2) - \lambda I \} = (-\frac{12}{29} - \lambda)(-\lambda) = 0$$

$$\therefore \lambda_1 = 0, \lambda_2 = -\frac{12}{29}$$

$\therefore$ all $|\lambda| < 1$, system is stable

2. This question explores the effect of the control weight R on closed-loop MPC stability. Let

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and consider the control system

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= x(k) \end{aligned}$$

with one-dimensional input $u \in \mathbb{R}$, and two-dimensional output $y = x$ (i.e., the output matrix is $C = I$). A model predictive control scheme with horizon length $H = 1$ is to be applied to by solving the optimization problem

$$\begin{aligned} \mathbb{P}_1(x(k)) : \quad & \min_{\mathbf{u}(k)} V_1 = \sum_{i=0}^1 \|\hat{x}(k+i|k)\|^2 + R|u(k|k)|^2 \\ & \text{subject to } \hat{x}(k+1|k) = A\hat{x}(k|k) + Bu(k|k), \\ & \mathbf{u}(k) = \{u(k|k)\}, \\ & \hat{x}(k|k) = x(k). \end{aligned} \tag{1.1}$$

The aim of this MPC scheme (with $Q = I$, $R > 0$ and $u_{\text{ref}} = 0, r = 0$) is to stabilize the system.

- (i) Is the original uncontrolled system stable?
- (ii) Show that the MPC cost function $V_1(x(k))$ can be written

$$V_1(x(k)) = x(k)^\top (I + A)x(k) + 2uB^\top Ax(k) + (1 + R)u^2.$$

where $u = u(k|k)$.

- (iii) By directly differentiating V_1 , show that the solution to the MPC optimization problem $\mathbb{P}_1(x(k))$ is

$$u(k) = K_1 x(k), \quad \text{where } K_1 = \frac{1}{1+R} \begin{pmatrix} 1 & -2 \end{pmatrix}$$

That is, the MPC control law $\kappa : \mathbb{R}^2 \rightarrow \mathbb{R}$ is given by $\kappa(x) = K_1 x$.

- (iv) Show that the closed-loop MPC dynamics are $x(k+1) = (A + BK_1)x(k)$.
- (v) For what values of R are the closed-loop dynamics stable (you may want to calculate the eigenvalues numerically)? Discuss how the choice of R affects the behaviour of the closed-loop MPC dynamics.

$$(i) \det(A - \lambda I) = (1 - \lambda)(2 - \lambda) + 1 = \lambda^2 - 3\lambda + 3 = 0$$

$$\therefore \lambda = \frac{3 \pm \sqrt{9-12}}{2} = \frac{3}{2} \pm \frac{1}{2}\sqrt{3}i \quad \therefore |\lambda| = \sqrt{3} > 1 \quad \therefore \text{unstable}$$

$$(ii) V_1(x(k)) = x(k)^T Q x(k) + x(k+1)^T Q x(k+1) + u(k)^T R u(k)$$

$$= x(k)^T Q x(k) + [Ax(k) + Bu(k)]^T Q [Ax(k) + Bu(k)] + u(k)^T R u(k)$$

$$= x(k)^T (Q + A^T Q A) x(k) + 2u(k)^T B^T Q A x(k) + u(k)^T B^T Q B u(k) + u(k)^T R u(k)$$

$$\therefore u \in \mathbb{R}, Q = I$$

$$\therefore V_1(x(k)) = x(k)^T (I + A^T A) x(k) + 2u(k)^T B^T A x(k) + (1 + R) u(k)^2$$

$$(iii) \frac{\partial V_1}{\partial u} = 2B^T A x(k) + 2(1+R)u(k)$$

$$\text{when } u(k) = u^*(k), \frac{\partial V_1}{\partial u} = 0$$

$$\therefore u^*(k) = -\frac{B^T A}{1+R} x(k) = \frac{1}{1+R} [1 \ -2] x(k) = K_1 x(k)$$

$$(iv) x(k+1) = Ax + Bu^*(k) = (A + BK_1) x(k)$$

$$(v) \text{ let } r = \frac{1}{1+R}$$

$$\therefore R > 0 \quad \therefore 0 < r < 1$$

$$\therefore A + BK_1 = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} + r \begin{bmatrix} 0 \\ 1 \end{bmatrix} [1 \ -2] = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ r & -2r \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ r-1 & -2r+2 \end{bmatrix}$$

$$\therefore \det[(A + BK_1) - \lambda I] = (1 - \lambda)(-2r + 2 - \lambda) - (r - 1) = 0$$

$$\therefore \lambda^2 + (2r - 3)\lambda + (-3r + 3) = 0$$

$$\therefore \lambda = \frac{1}{2} [(3 - 2r) \pm \sqrt{4r^2 - 3}]$$

$$\text{for } \frac{\sqrt{3}}{2} \leq r < 1, |\lambda_1| = \frac{1}{2} [(3 - 2r) + \sqrt{4r^2 - 3}] \geq |\lambda_2| = \frac{1}{2} [(3 - 2r) - \sqrt{4r^2 - 3}]$$

$$\therefore \frac{3}{2} - \frac{\sqrt{3}}{2} \leq |\lambda_1| < 1 \quad \therefore \text{system is stable}$$

$$\text{for } 0 < r < \frac{\sqrt{3}}{2}, |\lambda| = \frac{1}{4} [(3 - 2r)^2 + (3 - 4r^2)] = -3r + 3$$

$$\text{to let } |\lambda| < 1, -3r + 3 < 1 \quad \therefore r > \frac{2}{3}$$

$$\therefore \text{system is stable when } \frac{2}{3} < r < 1$$

$$\therefore \text{system is stable when } 0 < R < \frac{1}{2} \quad \text{less control penalty} \rightarrow \text{more aggressive control} \rightarrow \text{stable}$$

3. This question looks at the effect of the horizon length H on closed-loop MPC stability. Let

$$A = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

where $a^2 + b^2 = 1$. Consider the control system

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= x(k) \end{aligned}$$

A model predictive control scheme with horizon length H , weights $Q = I, R = 1$ and references $u_{\text{ref}} = 0, r = 0$ is to be applied to stabilize the system. This involved solving the MPC optimization problem

$$\begin{aligned} \mathbb{P}_H(x(k)) : \quad \min_{\mathbf{u}(k)} \quad V_H &= \sum_{i=0}^H \|\hat{x}(k+i|k)\|^2 + \sum_{i=0}^{H-1} |u(k+i|k)|^2 \\ \text{subject to} \quad \hat{x}(k+i+1|k) &= A\hat{x}(k+i|k) + Bu(k+i|k), \\ \mathbf{u}(k) &= \{u(k|k), u(k+1|k), \dots, u(k+H-1|k)\}, \\ \hat{x}(k|k) &= x(k). \end{aligned} \quad (1.2)$$

- (i) Suppose that $H = 2$. Show that

$$\begin{pmatrix} \hat{x}(k|k) \\ \hat{x}(k+1|k) \\ \hat{x}(k+2|k) \end{pmatrix} = \underbrace{\begin{pmatrix} I \\ A \\ A^2 \end{pmatrix}}_M x(k) + \underbrace{\begin{pmatrix} 0 & 0 \\ B & 0 \\ AB & B \end{pmatrix}}_W \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad (1.3)$$

where $u_1 = u(k|k)$ and $u_2 = u(k+1|k)$.

- (ii) If W is the matrix indicated above, show that

$$I + W^\top W = \begin{pmatrix} 3 & a \\ a & 2 \end{pmatrix}.$$

- (iii) By writing equation (1.3) in the form $\hat{X} = Mx(k) + WU(k)$ show that the MPC cost function can be written

$$V_2(x(k)) = \|Mx(k) + WU(k)\|^2 + \|U(k)\|^2$$

and that

$$\frac{\partial V_2}{\partial U} = \begin{pmatrix} \frac{\partial V_2}{\partial u_1} \\ \frac{\partial V_2}{\partial u_2} \end{pmatrix} = 2(I + W^\top W)U(k) + 2U(k)^\top W^\top Mx(k).$$

(iv) Hence, show that the solution to the MPC optimization problem $\mathbb{P}_2(x(k))$ is

$$U^*(k) = \begin{pmatrix} u^*(k|k) \\ u^*(k+1|k) \end{pmatrix} = -(I + W^\top W)^{-1} W^\top M x(k)$$

and that the MPC feedback law $\kappa : x(k) \mapsto u(k)$ is given by

$$\kappa(x) = - \begin{pmatrix} 1 & 0 \end{pmatrix} (I + W^\top W)^{-1} W^\top M x$$

(v) Verify numerically that the MPC feedback law in the case when $a = 0.9$ is given by

$$\kappa(x) = K_2 x \approx \begin{pmatrix} 0.20 & -0.59 \end{pmatrix} x$$

What are the associated closed-loop MPC dynamics? Are they stable?

(vi) We showed in lectures that the solution to the MPC problem $\mathbb{P}_H(x(k))$ for *any* horizon length H as the general form

$$U^*(k) = -(I + W^\top W)^{-1} W^\top M x(k) \quad (1.4)$$

where W, M and $U(k)$ are given by the expressions

$$M = \begin{pmatrix} I \\ A \\ A^2 \\ \vdots \\ A^H \end{pmatrix}, \quad W = \begin{pmatrix} 0 & 0 & \cdots & \cdots & 0 \\ B & 0 & \cdots & \cdots & 0 \\ AB & B & 0 & \cdots & 0 \\ A^2B & AB & B & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & 0 \\ A^{H-1}B & A^{H-2}B & A^{H-3}B & \cdots & B \end{pmatrix}, \quad U(k) = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_H \end{pmatrix},$$

and $u_1 = u(k|k), u_2 = u(k+1|k), \dots, u_H = u(k+H-1|k)$.

Write MATLAB code which can calculate the solution to (1.4) and also the MPC feedback law

$$u = K_H x = - \begin{pmatrix} 1 & 0 & \cdots & 0 \end{pmatrix} (I + W^\top W)^{-1} W^\top M x$$

for any horizon length H .

- (vii) Use MATLAB's in-built `d1qr` command to solve the discrete-time LQR problem with weighting matrices $Q = I$ and $R = 1$ in the case that $a = 0.9$.
- (viii) How does the LQR feedback gain K_{LQR} compare with your solution K_H from (vi) as the horizon length H increases?

$$(i) \hat{x}(k|k) = x(k)$$

$$\hat{x}(k+1|k) = Ax(k) + Bu_1$$

$$\hat{x}(k+2|k) = A\hat{x}(k+1|k) + Bu_2 = A^2x(k) + ABu_1 + Bu_2$$

$$(ii) W^T W = \begin{bmatrix} 0 & B^T & B^T A^T \\ 0 & 0 & B^T \end{bmatrix} \begin{bmatrix} 0 & 0 \\ B & 0 \\ AB & B \end{bmatrix} = \begin{bmatrix} B^T B + B^T A^T A B & B^T A^T B \\ B^T A B & B^T B \end{bmatrix}$$

where,

$$B^T B = 1,$$

$$B^T A^T A B = [0 \ 1] \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = [b \ a] \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = b^2 + a^2 = 1$$

$$B^T A B = [0 \ 1] \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = [-b \ a] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = a$$

$$B^T A^T B = [0 \ 1] \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = [b \ a] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = a$$

$$\therefore I + W^T W = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1+1 & a \\ a & 1 \end{bmatrix} = \begin{bmatrix} 3 & a \\ a & 2 \end{bmatrix}$$

$$\begin{aligned} (iii) Y_2(x(k)) &= x(k)^T Q x(k) + x(k+1)^T Q x(k+1) + x(k+2)^T Q x(k+2) + u_1^T R u_1 + u_2^T R u_2 \\ &= x(k)^T Q x(k) + [Ax(k) + Bu_1]^T Q [Ax(k) + Bu_1] \\ &\quad + [A^2x(k) + ABu_1 + Bu_2]^T Q [A^2x(k) + ABu_1 + Bu_2] + u_1^T R u_1 + u_2^T R u_2 \\ &= x(k)^T [Q + A^T Q A + (A^2)^T Q A^2] x(k) \\ &\quad + 2u_1^T (B^T Q A + B^T A^T Q A^2) x(k) + 2u_2^T B^T Q A^2 x(k) \\ &\quad + u_1^T (B^T Q B + B^T A^T Q A B) u_1 + 2u_1^T B^T A^T Q B u_2 + u_2^T B^T Q B u_2 \\ &\quad + u_1^T R u_1 + u_2^T R u_2 \\ &= [Mx(k)]^T Q [Mx(k)] + 2[WU(k)]^T Q [Mx(k)] + [WU(k)]^T Q [WU(k)] \\ &\quad + [U(k)]^T R [U(k)] \\ &= \|Mx(k) + WU(k)\|_Q^2 + \|U(k)\|_R^2 \end{aligned}$$

$$\therefore \frac{\partial Y_2}{\partial U} = 2W^T Q Mx(k) + 2W^T Q WU(k) + 2RU(k)$$

$$= 2(I + W^T W)U(k) + 2W^T Mx(k) \text{ for } Q = I, R = 1$$

(iv) at $U(k) = U^*(k)$, $\frac{\partial V_k}{\partial U} = 0$

$$\therefore U^*(k) = -(I + W^T W)^{-1} W^T M x(k) = \begin{bmatrix} u^*(k) \\ u^*(k+1) \end{bmatrix}$$

$$\therefore K(x) = B u^*(k) = [1 \ 0] U^*(k)$$

(v) $A + BK_k = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} [0.20 \ -0.59]$

$$= \begin{bmatrix} 0.9 & b \\ -b & 0.9 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0.20 & -0.59 \end{bmatrix}$$

$$= \begin{bmatrix} 0.90 & b \\ 0.20-b & 0.31 \end{bmatrix}$$

$$\therefore \det[(A + BK_k) - \lambda I] = \lambda^2 - 1.21\lambda + 0.279 - 0.2b + b^2 \quad \therefore \lambda = 0.605 \pm \frac{\sqrt{0.3481 + 0.8b - 4b^2}}{2}$$

$$\therefore a^2 + b^2 = 1 \quad \therefore b = \pm \frac{\sqrt{19}}{10}$$

$$\text{if } b = \frac{\sqrt{19}}{10}, \quad \lambda = 0.605 \pm 0.126i \quad \therefore |\lambda| = 0.618 < 1$$

$$\text{if } b = -\frac{\sqrt{19}}{10}, \quad \lambda = 0.605 \pm 0.436i \quad \therefore |\lambda| = 0.746 < 1$$

$\therefore$ the closed loop dynamics are stable

(vii) see MATLAB

(vi) see MATLAB

(v) K_{LAR} value is $[0.0232 \ -0.7701]$ in MATLAB $K = -K$

$K_H \rightarrow K_{\text{LAR}}$ as $H \rightarrow \infty$

4. Consider the control system

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\y(k) &= x(k)\end{aligned}$$

and the model predictive control scheme with horizon length $H = 1$:

$$\begin{aligned}\mathbb{P}_1(x(k)) : \quad & \min_{\mathbf{u}(k)} V_1 = \sum_{i=0}^1 \|\hat{x}(i|k)\|^2 + R|u(k|k)|^2 \\ \text{subject to} \quad & \hat{x}(k+1|k) = A\hat{x}(k|k) + Bu(k|k), \\ & \mathbf{u}(k) = \{u(k|k)\}, \\ & \hat{x}(k|k) = x(k).\end{aligned} \tag{1.5}$$

For each of the following choices of matrices A and B , calculate from first principles MPC feedback law $u = K_1 x$. In each case, determine if the closed-loop MPC dynamics are asymptotically stable. Note that in all cases it is assumed that the state weighting matrix is $Q = 1$ and the output matrix is $C = I$.

(i)

$$A = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad R = 1.$$

(ii)

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad R = 2.$$

(iii)

$$A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad R = 1.$$

(iv) In any case where you find the closed-loop dynamics $x(k+1) = (A + BK_1)x(k)$ are not asymptotically stable, investigate how increasing the horizon length affects the stability of the system. You may need to use your code from Q3 to do this.

for ease of writing, let $x = x(k)$, $u = u(k)$

$$\begin{aligned} V_1 &= x^T Q x + u^T R u + (Ax + Bu)^T Q (Ax + Bu) \\ &= x^T Q x + u^T R u + x^T A^T Q A x + 2u^T B^T Q A x + u^T B^T Q B u \\ &= x^T (I + A^T A) x + 2u^T B^T A x + (B^T B + R) u^T \quad \text{since } Q = I \end{aligned}$$

$$\therefore \text{ at } u = u^*, \frac{dV_1}{du^*} = 2B^T A x + 2(B^T B + R) u^* = 0$$

$$\therefore u^* = -(B^T B + R)^{-1} B^T A x = K x$$

$$\text{ii) } K = -(2+1)^{-1} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{6} & -\frac{1}{6} \end{bmatrix}$$

$$\therefore \det \{ (A + BK) - \lambda I \} = \det \left(\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} + \begin{bmatrix} -\frac{1}{6} & -\frac{1}{6} \\ -\frac{1}{6} & -\frac{1}{6} \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = \begin{vmatrix} \frac{1}{3} - \lambda & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} - \lambda \end{vmatrix} = 0$$

$$\therefore \lambda^2 - \frac{2}{3}\lambda + \frac{1}{9} - \frac{1}{18} = \lambda^2 - \frac{2}{3}\lambda + \frac{1}{18} = 0 \quad \therefore 18\lambda^2 - 12\lambda + 1 = 0 \quad \therefore \lambda_{1,2} = \frac{1}{3} \pm \frac{1}{6}\sqrt{2}$$

$$\therefore |\lambda_1| < |\lambda_2| < 1 \quad \therefore \text{stable}$$

$$\text{iii) } K = -(1+2)^{-1} \begin{bmatrix} 1 & 0 \end{bmatrix} = -\begin{bmatrix} \frac{1}{3} & 0 \end{bmatrix}$$

$$\therefore \det \{ (A + BK) - \lambda I \} = \det \left(\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} -\frac{1}{3} & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = \begin{vmatrix} \frac{2}{3} - \lambda & 0 \\ 1 - \lambda & 1 - \lambda \end{vmatrix} = 0$$

$$\therefore (\frac{2}{3} - \lambda)(1 - \lambda) = 0 \quad \therefore \lambda_1 = \frac{2}{3}, \lambda_2 = 1 \quad \therefore \text{marginally stable}$$

$$\text{dii) } K = -(1+1)^{-1} \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix}$$

$$\therefore \det \{ (A + BK) - \lambda I \} = \det \left(\begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} + \begin{bmatrix} -\frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right) = 0$$

$$\therefore \begin{vmatrix} \frac{1}{2} - \lambda & \frac{1}{2} & 0 \\ -1 & 1 - \lambda & 0 \\ 0 & -1 & 1 - \lambda \end{vmatrix} = (1 - \lambda) \left[(\frac{1}{2} - \lambda)(1 - \lambda) + \frac{1}{2} \right] = (1 - \lambda) (\lambda^2 - \frac{3}{2}\lambda + 1) = 0$$

$$\therefore \lambda_1 = 1, \lambda_{2,3} = \frac{3}{4} \pm \frac{\sqrt{7}}{4}i \quad \therefore \text{marginally stable}$$

iv) increase the H makes the system become stable

5. For a system $\dot{x} = f(x, u)$ and $y = h(x)$ consider the MPC problem whose goal is to force the output y to track a chosen constant reference signal r :

$$\begin{aligned} \mathbb{P}_H(x(k)) : \quad & \min_{\mathbf{u}(k)} \sum_{i=0}^H \|\hat{y}(k+i|k) - r\|_Q^2 + \sum_{i=0}^{H-1} \|u(k+i+1|k) - u(k+i|k)\|_R^2 \\ \text{subject to} \quad & \hat{x}(k+i+1|k) = f(\hat{x}(k+i|k), u(k+i|k)), \\ & \mathbf{u}(k) = \{u(k|k), u(k+1|k), \dots, u(k+H-1|k)\}, \\ & \hat{y}(k+i|k) = h(\hat{x}(k+i|k)), \\ & \hat{x}(k|k) = x(k). \end{aligned}$$

What are the advantages and disadvantages of penalizing the control 'steps'

$$\|u(k+i+1|k) - u(k+i|k)\|^2$$

rather than the control 'effort' $\|u(k+i|k)\|^2$ that we discussed in lectures?

1. restricting the control rate is benefit for practical application, since there are typically limits on how fast an controller can respond
2. perfect tracking can be achieved without the need to calculate an appropriate " u_{ref} ", avoid to find an appropriate (r, x_r, u_{ref}) triplet
3. possible disadvantage is not prevent control growing unrealistic large without constraints

6. The state constraints

$$-\mathbf{c} \leq \hat{\mathbf{x}}(k+i|k) \leq \mathbf{c}, \quad i = 1, \dots, H \quad (1.6)$$

are to be added to an MPC optimization problem, where $\mathbf{c} \in \mathbb{R}^n$ is a fixed vector of positive numbers and the inequality (1.6) is understood to apply element-by-element.

Suppose that the system is linear, with the predicted states satisfying

$$\hat{\mathbf{x}}(k+i+1|k) = A\hat{\mathbf{x}}(k+i|k) + Bu(k+i|k).$$

- (i) In the cases $H = 1$ and $H = 2$, show that there exists a matrix G and a vector $\mathbf{g}$ such that (1.6) are equivalent to the linear inequality constraint

$$GU(k) \leq \mathbf{g}, \quad \text{where} \quad U(k) = \begin{pmatrix} u(k|k) \\ u(k+1|k) \\ \vdots \\ u(k+H-1|k) \end{pmatrix}.$$

Convince yourself that the same is true for any H .

- (ii) Discuss the fundamental difference, in terms of feasibility, of imposing state constraints rather than control constraints.

i) when $H=1$, $-c \leq \hat{x}(k+1|k) \leq c$

$$\therefore -c \leq Ax(k) + Bu(k|k) \leq c$$

$$\therefore \begin{bmatrix} B \\ -B \end{bmatrix} U(k) \leq \begin{bmatrix} c - Ax(k) \\ c + Ax(k) \end{bmatrix}$$

when $H=2$, $-c \leq \hat{x}(k+2|k) \leq c$

$$\therefore -c \leq A^2 x(k) + ABu(k|k) + Bu(k+1|k) \leq c$$

$$\therefore \begin{bmatrix} B & AB \\ -B & -AB \end{bmatrix} U(k) \leq \begin{bmatrix} c - A^2 x(k) \\ c + A^2 x(k) \end{bmatrix}$$

ii) control constraint feasibility is mainly a question of actuator limits, might require invariant sets to ensure recursive stability

state constraint feasibility depends on system, easy to make MPC problem infeasible while control constraint typically do not